

ChingEn Kung

PHD STUDENT

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Professional Summary

I am a PhD student at National Taiwan University. My research interests include deep learning, numerical simulations, and Carbon Capture and Storage. Currently working on neural operators and Transformer-based solvers to accelerate multiphysics simulation problems.

Research Experience

Research Assistance | 2023-2025

National Taiwan University

- Develop deep learning surrogate model that is 1000 times faster than numerical simulations.
- 3D large scale numerical modeling of multiphase darcy's flow.

Academic History

PhD student, National Taiwan University

2025-

NSTC Research Scholarship recipient

Relevant Courses:

- Machine Learning and Kernel Method (CommE5053) - A
- Parallel Programming (CSIE5213) - this semester

M.S., National Central University

2021-2023

Thesis: Thermo-hydro-mechanical-chemical simulations on the evolution of pore-fluid pressure and deformation mechanisms of the fault gouge at seismic slip rate

B.S., National Central University

2017-2021

Geoscience

Selected Publications

A High-Fidelity Surrogate for Multiphase Flow in Complex Faulted System Using Geometric-Aware Fourier Neural Operator. 2026/5/19-22 Interpore2026 Poster Presentation.